Safety Trade-Off: Escalation Rate vs. Silent Failure Rate
Silent Failure Rate (%) [Novel Attacks Mistaken as Benign]
Escalation Rate to Tier 2 (%)
0
10
20
30
40
50
60
70
40
50
60
70
80
90
Single-Tier
Single-Tier
RF
Calibrated
Isolation
Split-Conformal
Bagged
Evidential
EDL
Single-Tier RF (No Escalation)
Single-Tier XGBoost (No Escalation)
RF Prob Band [0.30, 0.70]
Calibrated RF (Isotonic)
Bagged RF Tree Variance
Split-Conformal Prediction
Isolation Forest Novelty
Evidential Deep Learning (EDL)
EDL Ensemble (Warm-Start Prior)